

Q My baby's heart was beating on an early ultrasound! Has it formed yet?

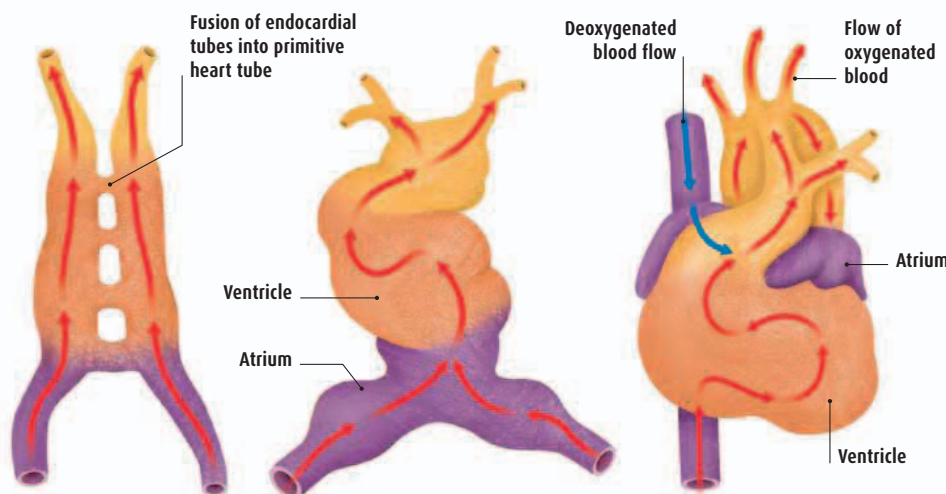
At only six weeks pregnant, your baby has a simple heart that's beating, and is doing it at up to 160 times per minute.

Your baby needs a circulation system so that nutrients can pass into all her cells as they develop. At first, the yolk sac and placenta can diffuse nutrients and oxygen into her cells, but pretty soon the embryo is too large for this method and she needs a circulation system to help. That's why the heart is one of the first organs a baby develops—as early as 18 days after fertilization.

The heart begins life as two cords (the endocardial tubes) that fuse together, creating a “primitive heart tube.” Once the heart tube has formed at around six weeks of pregnancy, it begins to pump blood cells. A mere three

weeks later, this tiny, early heart has developed its chambers and valves.

Since your baby doesn't breathe in the uterus, she gets her oxygen from her mother through the placenta. Blood that leaves the body would normally go to the lungs to collect oxygen, but a fetus's heart has two extra connections that allow deoxygenated blood to bypass the lungs. Instead, fetal blood vessels pass out through the umbilical cord to the placenta to pick up oxygen. On the return journey, the rerouting sends oxygenated blood to the brain cells quickly. After birth, the two extra connections in the heart close.



1 Primitive heart tube The two cords merge from the base upward, forming a single primitive heart tube about 21 days after fertilization (five weeks pregnant).

2 Bending of the heart tube After forming, the heart tube begins to beat. It then elongates and loops to the right, forming a spiral and a basic circulation.

3 Final position of chambers The tube twists again to create four chambers. By nine weeks, the chambers are totally separate, and tiny valves have formed inside.

Q Is there anything I can do to encourage healthy organ development in my baby?

Yes, there are certain ways you can help.

The first 10 weeks of pregnancy are the most vulnerable time for your baby's development, since it's when her organs are most susceptible to environmental harm. If your pregnancy is planned, you may already be taking folic acid

(to encourage healthy nerve and brain development in your baby) and be avoiding toxins. But if that isn't the case for you, don't worry. As soon as you find out you're pregnant, start taking supplements of folic acid (400mcg) and vitamin D (10mcg) daily; cut out alcohol, nicotine, and any other recreational drugs if you take them (seek advice if need be); and avoid passive smoking. You can also steer clear of certain foods to lower the risk of getting harmful food poisoning (see pp.52–53).

Q What is the neural tube, and how does it relate to my baby's brain and nervous system?

The neural tube is the structure that eventually becomes your baby's brain and nervous system.

It develops when you are around four weeks pregnant and, at around five weeks of pregnancy, it separates to form cells that will become the outer layer of your baby's brain (the cerebellum) and other cells that will travel throughout your baby's developing body to become nerves. You may have heard of the neural tube in relation to “neural tube defects.” These are conditions that can very rarely occur as a result of low maternal folic acid, among other reasons. You may be offered very accurate tests for these defects as part of your prenatal care (see p.97).

Q Why is my baby surrounded by amniotic fluid? What is it for?

Amniotic fluid protects your baby from trauma, aids lung development, helps maintain a constant temperature (slightly higher than your own), and gives your baby space to grow in. At the beginning of your pregnancy amniotic fluid is primarily made up of water and electrolytes. Around weeks 12–14 the fluid contains carbohydrates, proteins, lipids, phospholipids, and urea. As the kidneys start to function and produce urine, it passes into the amniotic fluid. By the end of your pregnancy your baby will swallow 1–2 pints (0.5–1 liters) of amniotic fluid a day and replace it as urine. The volume of amniotic fluid present in the amniotic sac decreases toward the end of the pregnancy as the fetal kidneys produce smaller amounts of more concentrated urine.



Did you know...

The volume of amniotic fluid at 12 weeks is about 1 fl oz (30 ml), less than an egg cup full. By 32 weeks the amniotic fluid in the sac reaches its highest volume. This is usually 2 pints (1 liter), although this can be as much as 4.25 pints (2 liters).